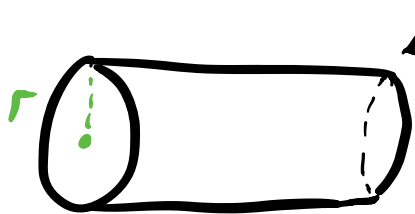


Volume of revolution

Basic skills

Q1.



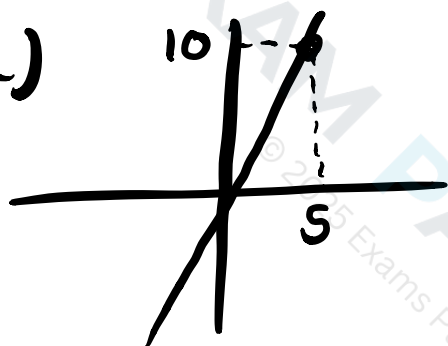
so $y = \text{constant}$

e.g. $y = 2$

radius 2 cylinder

Q2. $y = 2x$

a)



b) cone of height 5, radius 10

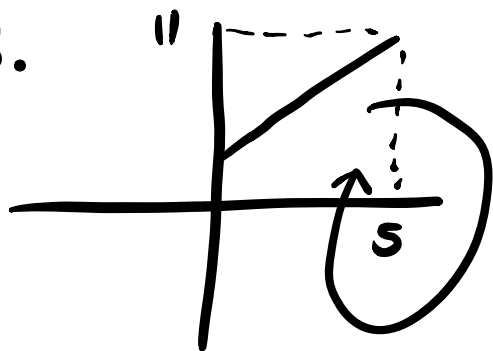
$$\begin{aligned} \text{c) } \int_0^5 \pi (2x)^2 dx &= 4\pi \int_0^5 x^2 dx = 4\pi \left[\frac{1}{3} x^3 \right]_0^5 \\ &= 4\pi \left(\frac{1}{3} \times 125 - 0 \right) = \frac{500}{3} \pi \end{aligned}$$

d) cone formula is $v = \pi r^2 \frac{h}{3}$

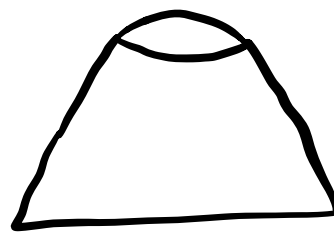
$$\text{so } v = \pi \times 10^2 \times \frac{5}{3} = \frac{500}{3} \pi$$

$$\begin{aligned} \text{e) } x = \frac{y}{2} \text{ so } \int_0^5 \pi \left(\frac{y}{2} \right)^2 dy &= \frac{\pi}{4} \int_0^5 y^2 dy = \frac{\pi}{4} \left[\frac{1}{3} y^3 \right]_0^5 \\ &= \frac{\pi}{4} \left(\frac{1}{3} \times 125 - 0 \right) = \frac{125}{12} \pi \text{ so NO} \end{aligned}$$

Q3.



Frustum



$$\begin{aligned} \int_0^5 \pi (2x+1)^2 dx &= \pi \int_0^5 4x^2 + 4x + 1 dx \\ &= \pi \left[\frac{4}{3}x^3 + 2x^2 + x \right]_0^5 = \pi \left(\frac{4}{3} \times 5^3 + 2 \times 5^2 + 5 - 0 \right) \\ &= \frac{665}{3} \pi \end{aligned}$$

Q4.

$$\begin{aligned} \int_0^{\ln 2} \pi (1 - e^{-x})^2 dx &= \pi \int_0^{\ln 2} (1 - 2e^{-x} + e^{-2x}) dx \\ &= \pi \left[x + 2e^{-x} - \frac{1}{2}e^{-2x} \right]_0^{\ln 2} \\ &= \pi \left(\ln 2 + 2e^{-\ln 2} - \frac{1}{2}e^{-2\ln 2} - (0 + 2e^0 - \frac{1}{2}e^0) \right) \\ &= \pi \left(\ln 2 + 1 - \frac{1}{8} - (0 + 2 - \frac{1}{2}) \right) \\ &= \left(\ln 2 - \frac{5}{8} \right) \pi \end{aligned}$$

Q5.

$$\pi \int_{-2}^0 (4 - 2x + x^3) dx$$

$$= \pi \int_{-2}^0 (16 - 16x + 4x^2 + 8x^3 - 4x^4 + x^6) dx$$

$$= \pi \left[16x - 8x^2 + \frac{4}{3}x^3 + 2x^4 - \frac{4}{5}x^5 + \frac{1}{6}x^6 \right]_{-2}^0$$

$$= \pi \left(0 - (16 \times -2 - 8(-2)^2 + \frac{4}{3}(-2)^3 + 2(-2)^4 - \frac{4}{5}(-2)^5 + \frac{1}{6}(-2)^6) \right)$$

$$= \pi \times \frac{32}{5} = \frac{32}{5} \pi \quad \text{so if } x \text{ is worth } 10 \text{ cm}$$

Actual volume is $\times 1000$ $\leftarrow 10^3$

$$\text{so } \frac{32}{5} \pi \times 1000 = 6400 \pi$$

Q6. $x = y^{\frac{1}{a}}$

$$\text{so } \pi \int_0^1 (y^{\frac{1}{a}})^2 dy = 1 \Rightarrow \int_0^1 y^{\frac{2}{a}} dy = \frac{1}{\pi}$$

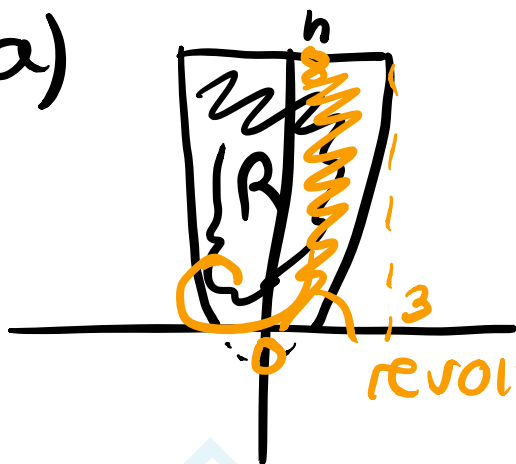
$$\Rightarrow \left[y^{\frac{2+a}{a}} \times \frac{a}{2+a} \right]_0^1 = \frac{1}{\pi} \Rightarrow \frac{a}{2+a} - 0 = \frac{1}{\pi}$$

$$\Rightarrow a\pi = 2+a \Rightarrow a(\pi-1) = 2$$

$$\text{so } a = \frac{2}{\pi-1}$$

Q7. $y = x^2 - 1$

a)



$$h = 8$$

$$\text{so } x^2 - 1 = 8 \quad x^2 = 9$$

$$x = 3 \text{ or } -3$$

$$\text{so } x^2 = y + 1 \quad x = \pm \sqrt{y + 1}$$

$$\text{only need one so } x = \sqrt{y + 1}$$

$$\text{Then } \pi \int_0^8 (\sqrt{y + 1})^2 dy = \pi \int_0^8 y + 1 dy$$

$$= \pi \left[\frac{1}{2} y^2 + y \right]_0^8 = \pi \left(\frac{1}{2} 8^2 + 8 - 0 \right) = 40\pi$$

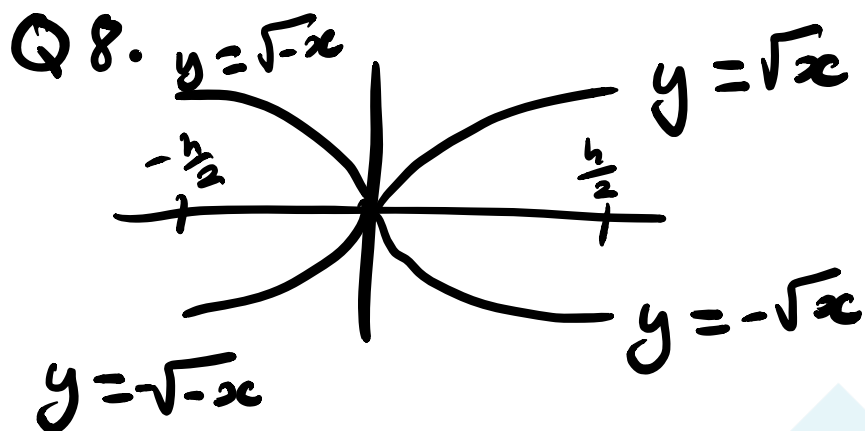
$$\text{b) } \pi \int_0^{h_1} y + 1 dy = 2\pi \int_0^{h_2} y + 1 dy$$

$$\text{so } \left[\frac{y^2}{2} + y \right]_0^{h_1} = 2 \left[\frac{y^2}{2} + y \right]_0^{h_2}$$

$$\Rightarrow \frac{h_1^2}{2} + h_1 = h_2^2 + 2h_2 \quad \text{is the relationship}$$

$$\text{e.g. if } h_2 = 8 \text{ then } \frac{h_1^2}{2} + h_1 = 80 \Rightarrow h_1^2 + 2h_1 - 160 = 0$$

$$h_1 \approx 11.688$$



As symmetric just $y = \sqrt{x}$ for $0, \frac{h}{2}$
revolved and doubled

$$\text{so } 2 \times \pi \int_0^{\frac{h}{2}} (\sqrt{x})^2 dx = 2\pi \left[\frac{1}{2} x^2 \right]_0^{\frac{h}{2}}$$

$$= \pi \left(\left(\frac{h}{2} \right)^2 - 0 \right) = \frac{\pi h^2}{4}$$

Q9.

a) $\pi \int_a^b \left(\frac{1}{x} \right)^2 dx = \pi \int_a^b x^{-2} dx$

As $\int x^{-2} dx = -\frac{1}{x}$ as $x \rightarrow 0$ goes to infinity

it will not integrate if

$[a, b]$ contains zero. so no

e.g. $a = -2, b = 4$ won't exist

b) $\pi \int_1^b \bar{x}^2 dx = \pi \left[-\frac{1}{\bar{x}} \right]_1^b = \pi \left(-\frac{1}{b} + 1 \right)$

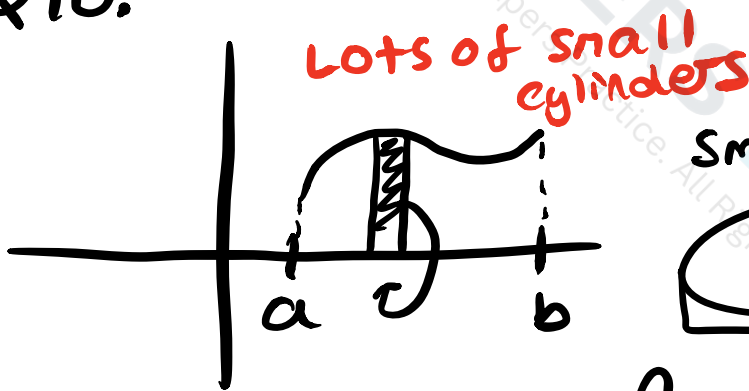
as $b \rightarrow \infty$ volume $\rightarrow \pi(0+1) = \pi$

c) The statement is wrong as seen in Part b).



Trumpet shape has finite vol, infinite length

Q10.



small volume $\delta V = \pi y^2 \delta L$



$\delta L = h$ ← small x change

with $n = \frac{b-a}{h}$

Then volume = $\sum_{i=1}^n \pi y(a+ih)^2 h$ Strips

as $n \rightarrow \infty$

volume = $\int_a^b \pi y^2 dx$

Sum becomes integral

































Advanced skills

